

GURU

Guru AI - SEE 2081 (2025) Model Solutions

Optional II Computer Science (RE-3031) - Full Solutions

Group 'A'

Ans 1.

- (a) Telecommunication is the transmission of information, signals, or data over significant distances by electronic or electromagnetic means.
 - (b) Two examples of simplex mode are Radio broadcasting and Television broadcasting.
 - (c) Two methods of creating a table in MS-Access are using 'Design View' and using 'Datasheet View'.
 - (d) The maximum character limit you can use for creating a field name in MS-Access is 64 characters.
 - (e) The syntax of the KILL statement is: KILL "filename.ext"
 - (f) Two keywords used in C-language are `int` and `return`.
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Ans 2.

- (a) Search Engine
- (b) Digital Footprint

Ans 3.

- (a) UPS: Uninterruptible Power Supply
 - (b) VR: Virtual Reality
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Group 'B'

Ans 4.

(a) Guided vs Unguided Media:

- 1. Guided media uses physical cables (like fiber optics, twisted pair) to transmit data, whereas unguided media transmits data through the air or vacuum using electromagnetic waves (like radio waves, microwaves).*
- 2. Guided media generally provides higher security and faster transmission speeds compared to unguided media.*

(b) Cyber Law & Cyber Ethics:

Cyber law is the area of law that deals with the internet, cyberspace, and their respective legal issues to prevent cybercrimes.

Do: Always use strong, unique passwords for your online accounts.

Don't: Do not engage in cyberbullying or harassment of others online.

(c) Antivirus Software:

Antivirus software is a computer program designed to detect, prevent, search for, and remove malicious software (viruses, worms, trojans) from a computer system.

Examples: Kaspersky, Avast.

(d) E-commerce vs Traditional Commerce:

- 1. E-commerce allows businesses to operate 24/7 without geographic limitations, making shopping highly convenient for consumers.*
- 2. It reduces operational costs (like physical store maintenance) and provides customers with a wider variety of choices and easy price comparisons.*

(e) E-governance:

E-governance is the application of information and communication technology (ICT) for delivering government services, exchange of information, and integration of various systems between government and citizens.

Examples in Nepal: NagariK App, Online Passport Application System.

(f) Database:

A database is an organized collection of structured information or data, typically stored electronically in a computer system, allowing for easy access, management, and updating.

Two data types in MS-Access: Short Text, Number.

(g) Field vs Record:

- 1. A field represents a single piece of specific information (a column in a table), while a record is a complete set of related fields for a specific entity (a row in a table).*
- 2. Example: 'Name' or 'Age' is a field, whereas the complete details of 'Ram' (Name: Ram, Age: 16) is a record.*

(h) Report in DBMS:

A report is a database object used to summarize, format, and present data from tables or queries in a printable and easy-to-read format. It is necessary in DBMS because it allows users to analyze data effectively for decision-making and to share the organized information with others.

(i) Query:

A query is a request for data or information from a database table or combination of tables based on specific criteria or conditions.

Types of Action queries: Make-Table Query, Append Query, Update Query, Delete Query.

Ans 5. (Dry Run & Output)

T\$	C	D\$	PRINT D\$;
"COMPUTER"	1	"C"	C
	3	"M"	M
	5	"U"	U
	7	"E"	E
	9	Loop Terminates	

Output: CMUE

Ans 6. (Debugging QBASIC)

```

REM to add record in an existing file.
OPEN "student.dat" FOR APPEND AS #2 'Correction 1: OUT changed to
APPEND
TOP:
INPUT "Enter Name, Class and Roll No."; SName$, C, RN
WRITE #2, SName$, C, RN 'Correction 2: INPUT#2 changed to
WRITE #2, variable SNames$ fixed
INPUT "More records"; Y$
IF UCASE$(Y$) = "Y" THEN GOTO TOP 'Correction 3: POP changed to TOP
CLOSE #2
END 'Correction 4: STOP changed to
END

```

Ans 7. (Program Analysis)

(a) Missing statement: To execute the function and print the returned value, the missing statement is:

`PRINT "Total Vowels = "; COUNT(W$)`

(b) String functions used: The two string functions used in the given program are `LEN( )` and `MID$( )`.

Group 'C'

Ans 8. (Number System)

i) $(1503)_8 = (?)_{16}$

Octal to Binary: $1 = 001, 5 = 101, 0 = 000, 3 = 011 \rightarrow (001101000011)_2$

Binary to Hex (groups of 4): $0011 \mid 0100 \mid 0011 \rightarrow 3 \mid 4 \mid 3$

Answer: $(343)_{16}$

ii) $(101000101)_2 = (?)_8$

Group binary into 3s from right to left: $101 \mid 000 \mid 101$

Convert to Octal: $5 \mid 0 \mid 5$

Answer: $(505)_8$

iii) $(1010 + 1101)_2 - (110)_2$

Addition: $1010 + 1101 = 10111$

Subtraction: $10111 - 110 = 10001$

Answer: $(10001)_2$

iv) $(100111)_2 \div (110)_2$

Quotient: 110

Remainder: 11

Answer: Quotient = 110 , Remainder = 11

Ans 9. (QBASIC Programming)

(a) QBASIC Program (Function & Sub Procedure)

```
DECLARE FUNCTION Peri (L, B)
DECLARE SUB Vol (L, B, H)

CLS
INPUT "Enter Length of room: ", L
INPUT "Enter Breadth of room: ", B
INPUT "Enter Height of room: ", H

PRINT "Perimeter of room = "; Peri(L, B)
CALL Vol(L, B, H)
END

FUNCTION Peri (L, B)
    Peri = 2 * (L + B)
END FUNCTION

SUB Vol (L, B, H)
    V = L * B * H
    PRINT "Volume of room = "; V
END SUB
```

(b) QBASIC Program (Sequential File)

```
OPEN "std.dat" FOR INPUT AS #1
CLS
PRINT "Students scoring more than 40 in Computer:"
PRINT "Name", "Roll No", "English", "Nepali", "Maths", "Computer"
PRINT "-----"

WHILE NOT EOF(1)
    INPUT #1, N$, R, E, Ne, M, C
    IF C > 40 THEN
        PRINT N$, R, E, Ne, M, C
    END IF
WEND

CLOSE #1
END
```

Ans 10. (C Programming)

Option 1: Greatest among two numbers

```
#include <stdio.h>

int main() {
    int a, b;
    printf("Enter two different numbers: ");
    scanf("%d %d", &a, &b);

    if(a > b) {
        printf("The greatest number is %d", a);
    } else {
        printf("The greatest number is %d", b);
    }
    return 0;
}
```

Option 2: OR (Display series up to 15th term)

```
#include <stdio.h>

int main() {
    int i, a = 5;
    printf("The series up to the 15th term is:\n");

    for(i = 1; i <= 15; i++) {
        printf("%d ", a);
        a = a + 5;
    }
    return 0;
}
```